

# Harshil Bhatia

(412) 579-1985 [linkedin.com/in/harshil-bhatia](https://www.linkedin.com/in/harshil-bhatia) [Google Scholar harshilb@andrew.cmu.edu](https://scholar.google.com/citations?user=harshilb@andrew.cmu.edu)

## EDUCATION

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|----------------------------------------------------------------------------------------------|---------------------------|
| Carnegie Mellon University                                                                   | Pittsburgh, PA   Dec 2026 |
| <i>Masters in Computer Vision</i>                                                            | GPA: 4.22/4.0             |
| Coursework: Advanced Computer Vision, Reinforcement Learning, Learning for 3D Vision.        |                           |
| Indian Institute of Technology, Jodhpur                                                      | Jodhpur, India   May 2023 |
| <i>Bachelors in Computer Science</i>                                                         | GPA: 8.5/10               |
| Coursework: Computer Graphics, Autonomous Systems, Pattern Recognition and Machine Learning. |                           |

## SKILLS

**Languages & Frameworks:** Python (pytorch, keras, numpy, pandas) C++, CUDA, Jupyter, Git, AWS, Docker, Linux, Conda, SQL.

**Learning and Vision:** Deep Learning, Generative AI (Diffusion Models, GANs), Transformers, Vision Transformers (ViTs), Self-supervised Learning, Reinforced Learning.

## EXPERIENCE

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| <b>Avataar.ai</b>   <i>Research Engineer-II</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Bangalore, India   Jun 2023 - Jun 2025.     |
| <ul style="list-style-type: none"><li>Engineered a real-time, agent-based LLM system for personalized product video generation; reduced generation latency by 10x via asynchronous workflows and optimized GPU inference.</li><li>Delivered real-time 3D reconstruction with Gaussian Splatting at 120+ FPS under 20 MB memory, optimizing latency-critical CUDA pipelines.</li><li>Led the development of an end-to-end automated product photography system using diffusion models; designed scalable modules for quantization, relighting, and client-specific fine-tuning – showcased by <a href="#">AWS</a>.</li></ul> |                                             |
| <b>Max Planck Institute for Informatics</b>   <i>Research Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Saarbrücken, Germany   Aug 2021 – Aug 2022. |
| <u>Advisor:</u> Dr. Vladislav Golyanik <ul style="list-style-type: none"><li>Formulated finding cyclic consistency matchings for three deformable isometric shapes.</li><li>Extrapolated the optimization to encompass a generalized shape scenario, exhibiting linear scalability.</li><li>Established new quantum SOTA and achieved results comparable with classical counterparts [1].</li></ul>                                                                                                                                                                                                                         |                                             |

## RESEARCH PROJECTS

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| <b>Matryoshka style Elastic Attention</b>   <i>Carnegie Mellon University</i>                                                                                                                                                                                                                                                                                                                                                                                                       | PA   Oct 2025 – Present.  |
| <u>Advisor:</u> Dr. Albert Gu and Prof. Zico Kolter <ul style="list-style-type: none"><li>Worked on developing nested attention modules solving a sparse KV cache problem using Multi-head Latent Attention.</li><li>Developed an efficient token-level routing mechanism inspired from Mixture of Experts models.</li><li>Designed the architecture and implemented the complete pre-training pipeline.</li></ul>                                                                  |                           |
| <b>Shape completion using flow Matching</b>   <i>Carnegie Mellon University</i>                                                                                                                                                                                                                                                                                                                                                                                                     | PA   Sept 2023 – Present. |
| <u>Advisor:</u> Dr. Jeff Ichnowski <ul style="list-style-type: none"><li>Developed a flow matching based model for novel view depth synthesis.</li><li>Formulated a new confidence aware flow loss obeying Bregmann Divergences.</li></ul>                                                                                                                                                                                                                                          |                           |
| <b>Conditional Radiance Field for Scene Forecasting</b>   <i>UC Berkeley</i>                                                                                                                                                                                                                                                                                                                                                                                                        | CA   Aug 2023 – Mar 2024. |
| <u>Advisor:</u> Prof. Joseph Gonzalez <ul style="list-style-type: none"><li>Developed CARFF, a 3D scene forecasting method using PC-VAE and NeRF, achieving 28+ PSNR in probabilistic modeling for occluded autonomous driving scenarios in CARLA simulator.</li><li>Engineered a two-stage training pipeline and Mixture Density Network, resulting in 100% collision avoidance success over 30 trials in multi-agent planning under uncertainty. Published in ECCV [2].</li></ul> |                           |

## SELECTED PUBLICATIONS

- [1] CCQuantuMM: Cycle-Consistent Quantum-Hybrid Matching of Multiple Shapes in CVPR, 2023 - **H. Bhatia**, E. Tretschk, Z. Lähner, M.S. Benkner, M. Moeller, C. Theobalt and V. Golyanik.
- [2] CARFF: Conditional Auto-encoded Radiance Field for 3D Scene Forecasting in ECCV, 2024 - J. Yang, K. Desai, C. Packer, **H. Bhatia**, N. Rhinehart, R. McAllister, J. Gonzalez.
- [3] SynthProv: Interpretable Framework for Profiling Identity Leakage in WACV, 2024 - J. Singh\*, **H. Bhatia\***, A. Bharati, R. Singh and M. Vatsa.

## AWARDS & VOLUNTEERING

- Published **8 peer-reviewed articles** with over **110 citations**, an h-index of 4, and an i10-index of 3 ([Full Pub list](#)).
- Top 0.1 percentile in IIT-JEE 2019 among 1.3 million applicants.
- A\* grade (exceptional performance) in Pattern Recognition and Machine Learning.
- Served as a reviewer for Neurips 2025, AAAI 2025.